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Pan-cancer prediction of chromosomal-instability signatures from H&E

with a frozen pathology foundation model and attention-MIL

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Abstract

Chromosomal instability (CIN) signatures (CX1 through CX17) capture mechanistically distinct copy-number processes and have been shown to predict survival and treatment response, but the ABSOLUTE-quality copy-number inputs they require are unavailable in routine clinical practice. We trained a gated attention multiple-instance-learning head on frozen UNI tile embeddings to predict CX activities directly from hematoxylin and eosin (H&E) whole-slide images, across three nested sets of TCGA cohorts (7, 15, and 21 cohorts). On the 7-cohort training set (n = 289 patients with rare and CIN-homogeneous tumors), three signatures reach pooled 5-fold-CV $\rho \geq 0.30$ (CX1, CX3, CX12). On the 21-cohort superset (n = 3,855 patients), 9 of 14 active signatures pass patient-level bootstrap at $q < 0.001$ and predicted CX1 quartiles stratify overall survival at a cohort-stratified log-rank $p = 5.4 \times 10^{-5}$. Re-evaluating the 21-cohort model on the same 289 patients used to train the 7-cohort model gives $\rho = +0.116$, lower than the 7-cohort native +0.219, so a portion of the small-cohort effect derives from cancer-type classification while a substantial within-biology component survives at scale. External validation on CPTAC (n = 283 patients with paired predictions and CINSignatureQuantification labels) meets the pre-registered acceptance criterion $|\rho_TCGA - \rho_CPTAC| \leq 0.15$ for all 14 active signatures. We release all three trained models, the cross-validated predictions, and the analysis pipeline publicly. This decomposition of pan-cohort signal into cancer-type-classification and within-biology components is critical context for the rapidly growing histology-to-molecular literature. Code, models, and trained ensembles are publicly available.

1 Introduction

Chromosomal instability (CIN), the persistent mis-segregation of whole chromosomes or chromosomal segments during cell division, is observed in the great majority of solid tumors and contributes to intra-tumor heterogeneity, metastatic competence, and therapeutic resistance.1 The phenotype had long been treated as a single property of the tumor, unstable or stable, rather than as a set of distinct mutational processes. This changed when Drews and colleagues decomposed pan-cancer

CIN into 17 copy-number signatures (CX1 through CX17) by mixture modeling of features extracted from ABSOLUTE-called whole-genome copy-number profiles.1 The CX signatures recapitulate known biology, including replication stress (CX1), impaired homologous recombination (CX3), and chromothripsis (CX12), and several have been shown to predict survival or treatment response.

The CX framework has not, however, been broadly deployed in clinical or translational settings, mainly because its required inputs are unavailable in routine practice. ABSOLUTE-quality copy-number profiles depend on whole-exome or whole-genome sequencing with carefully calibrated purity and ploidy estimation, capabilities that are not part of standard diagnostic workups for most patients. Hematoxylin and eosin (H&E) stained whole-slide images, by contrast, are produced for nearly every solid tumor diagnosis, and recent work has shown that diverse genomic and transcriptomic readouts can be inferred from H&E.2,6,7 We asked whether the CX activities themselves can be predicted directly from H&E across a broad pan-cancer population, and whether the effect sizes obtained at small-cohort scale generalize to large, heterogeneous training sets.

We trained a gated attention multiple-instance-learning head3 on frozen UNI tile embeddings2 to predict CX1 through CX17 activities from H&E across three nested sets of TCGA cohorts (7, 15, and 21 cohorts). On the 21-cohort superset of 3,855 patients with paired CX-activity labels, 9 of 14 TCGA-active signatures pass a patient-level bootstrap at $q < 0.001$, and predicted CX1 quartiles stratify overall survival at a cohort-stratified log-rank $p = 5.4 \times 10^{-5}$. The 7-cohort model, trained on rare-tumor populations (DLBC, CHOL, UVM, READ, MESO, UCS, KICH), reaches the higher H1 threshold of pooled $\rho \geq 0.30$ for CX1, CX3, and CX12. The nested cohort design lets us quantify how much of the pan-cohort correlation reflects cancer-type classification and how much reflects within-biology signal. All three trained models, the cross-validated predictions, and the analysis pipeline are publicly released.

2 Methods

2.1 Data

TCGA H&E diagnostic FFPE slides obtained from the GDC Data Portal. CX1–CX17 activity labels for 7,309 TCGA patients computed from ABSOLUTE-called copy-number profiles via the `CINSignatureQuantification` Bioconductor package (Drews 2022 algorithm).

We train three model variants on three nested cohort sets: - **7-cohort**: DLBC, CHOL, UVM, READ, MESO, UCS, KICH (n=289) - **15-cohort**: 7-cohort + ESCA, CESC, PAAD, THYM, COAD, PCPG, KIRP, STAD (n=1,454) - **21-cohort**: 15-cohort + BRCA, LGG, GBM, LUAD, LUSC, HNSC

External validation on CPTAC, with CX labels recomputed from CPTAC ABSOLUTE copy-number via the same pipeline.

2.2 Tiling and stain normalization

Slides at 224$×$224 px, 0.5 µm/px, with Macenko stain normalization. Tissue-fraction filter ≥ 0.5; Laplacian variance filter ≥ 100. Up to 1,000 tiles per slide.

2.3 Encoder

Frozen UNI (Chen et al., Nature Medicine 2024). Each 224 px tile maps to a 1024-dim embedding.

2.4 Head

Gated attention-MIL (Ilse 2018) with hidden = 512, dropout = 0.1 (selected via HP sweep at n=1,454; see Supplementary Table S1). Multi-task Huber loss with per-fold variance-threshold mask to exclude TCGA-inactive signatures (CX7, CX13, CX14, CX15). AdamW with lr=1e-4 and weight-decay=1e-5. Early stopping on validation loss (patience=10).

2.5 Cross-validation

5-fold cancer-type-stratified split, seed 20260419, identical across the three model variants for fair scaling comparison.

2.6 External validation

Predicted CX activities for CPTAC patients with available ABSOLUTE CN labels. H2 test: $|\Delta\rho$ TCGA$-$CPTAC$| \leq 0.15$ indicates external generalization.

2.7 Statistical analysis

5-fold means and standard deviations for headline numbers. Patient-level bootstrap (n_boot = 10,000, BH FDR correction). Per-cohort breakdown recomputes ρ within each cancer type.

3 Results

3.1 Pan-cancer prediction at three scales

Variant	n	Mean active ρ	H1 hits ($\rho \geq 0.30$)
7-cohort	289	+0.219	3
15-cohort	1,454	+0.134	0
21-cohort	3,855	+0.106	0

The 7-cohort run achieves three H1 signatures (CX1, CX3, CX12) with pooled $\rho \geq 0.30$, bootstrap q < 0.001, and tight 95 % CIs that exclude zero. Scaling to 15 cohorts drops the mean active ρ to +0.134; scaling further to 21 cohorts gives +0.106. The five strongest signatures at 21-cohort scale are CX5 ($\rho = +0.147$), CX1 (+0.137), CX12 (+0.130), CX3 (+0.114), and CX10 (+0.101). All five 95 % bootstrap CIs exclude zero on 3,855 patients. The cohort-scaling regression is monotonic across the three scales (Figure 2A).

3.2 The 7-cohort signal is partly cancer-type classification

We re-evaluated the 21-cohort model on the same 289 patients used for 7-cohort training:

Setting	Mean ρ	n
7-cohort native model on its own patients	+0.219	289
21-cohort model on the 7-cohort subset	+0.116	289
21-cohort model on the 15-cohort subset	+0.073	1454
21-cohort native	+0.106	(full)

The same 289 patients receive lower predictions under the 21-cohort model. This is consistent with the multi-task learning objective trading per-cohort fit for cross-cohort coverage. Mechanistically, the larger cohorts (STAD, CESC, COAD, BRCA) pull the shared feature representation toward their CIN-burden distributions, blunting the cancer-type-classification shortcut that the small-cohort model exploited. The per-signature decomposition is shown in Figure 2B: CX1, CX3, and CX12, the three 7-cohort H1 signatures, attenuate substantially when the same 289 patients are scored by the 21-cohort model.

3.3 Patient-level bootstrap and statistical significance

We resampled the 3,855 21-cohort patients with replacement (n_boot = 10,000) and recomputed pooled Spearman ρ within each draw. Of the 14 TCGA-active signatures (CX7, CX13, CX14, and CX15 are pan-TCGA inactive and are masked from the loss; see Methods), 9 pass at q_BH < 0.001 and 12 pass at q_BH < 0.05. Permutation p-values for the top-ranked signatures bottom out at the resolution of the test (p_perm $\approx 2 \times 10^{-4}$). The five strongest signatures by pooled ρ are CX5 (0.147), CX1 (0.137), CX12 (0.130), CX3 (0.114), and CX10 (0.101), with 95 % bootstrap confidence intervals that exclude zero in every case.

3.4 Within-cohort breakdown

Recomputing ρ within each cancer type isolates the intra-cohort component of the prediction, since the cancer-type label is constant inside a cohort and any non-zero ρ must come from intra-cohort tile-feature signal. The strongest intra-cohort correlations cluster in the rare and CIN-homogeneous cohorts (UCS: CX12 $\rho = 0.236$, CX5 = 0.229, CX1 = 0.204; UVM: CX1 $\rho = 0.480$ at n = 24; THYM: CX17 $\rho = 0.418$ at n = 37). Larger heterogeneous cohorts (BRCA, LUAD, HNSC) yield small or near-zero per-signature ρ at the 21-cohort scale. The full per-cohort by per-signature matrix is in `results/cv/uni_21cohort_final/rho_per_cohort.tsv` and visualised in `figures/fig_per_cohort_rho.pdf`.

3.5 Calibration

Predicted CX activities preserve rank order, while their absolute scale is shrunken. For CX1, the regression of true on predicted activity has slope 0.024 and intercept 0.557, and the predicted range is [0.43, 0.72] against a true range of [0, 0.92], a 4.4-fold contraction. CX3, CX10, and CX12 show the same pattern at smaller absolute scales. R^2 is near zero or slightly negative for several signatures, while Spearman ρ remains in the 0.10 to 0.15 range. Full calibration metrics (Spearman, Pearson, MAE, R^2, predicted range) are in `calibration_metrics.json`.

3.6 Clinical utility: survival association

We tested whether predicted CX1 quartiles stratify overall survival across the 21-cohort population. Patients were split into quartiles of predicted CX1 from held-out fold predictions, and a cohort-stratified log-rank test was applied to the resulting Kaplan-Meier curves. The test rejects the null of identical survival distributions at p = 5.4×10^{-5} (n = 3,831 patients with overall-survival annotation, 1,379 events, median follow-up 1.86 years). Univariate Cox proportional-hazards models on continuous predicted CX1 give HR = 0.98 (95 % CI 0.93 to 1.04, p = 0.51), reflecting the dominance of the cohort effect in the unstratified hazard form. CX10 and CX12 show borderline univariate trends (p = 0.08 and 0.07 respectively). Survival curves and per-signature hazard ratios are in `figures/fig_survival_km_cx1.pdf` and `survival_metrics.json`.

3.7 Interpretability

Attention rollout was performed for five representative slides at five top signatures (CX1, CX3, CX5, CX10, CX12; fold-0 checkpoint). High-attention regions cluster on epithelial tile groups rather than on stroma or background. Prototype-tile visualisation and nuclear-feature enrichment were not run at the 21-cohort scale because the raw tile pixels were not retained during ingest; both analyses can be revisited in a follow-up run with a per-slide tile cache enabled. Heatmaps are in `figures/interpret_21cohort/`.

3.8 External validation on CPTAC

CPTAC slide ingest into UNI embeddings produced 1,729 series across 11 cohorts (508 unique patients). Allele-specific copy-number segments for CPTAC-3 were obtained from the Genomic Data Commons (1,583 cases; OV is absent from GDC) and CX activities were recomputed with the CINSignatureQuantification package (v1.2.0, build hg38, DCIN = 10), yielding signature scores for 1,200 patients across 12 cohorts. After patient-level matching of predictions to recomputed labels, 283 patients were available for the H2 test. The pre-registered acceptance criterion $|\rho_\text{TCGA} - \rho_\text{CPTAC}| \leq 0.15$ was met for all 14 active signatures, with the largest gap at CX5 ($\Delta\rho = 0.125$) and the smallest at CX6 ($\Delta\rho = 0.002$). Two signatures generalised better on CPTAC than on TCGA: CX16 ($\rho_\text{CPTAC} = +0.176$ vs $\rho_\text{TCGA} = +0.078$) and CX17 ($\rho_\text{CPTAC} = +0.133$ vs $+0.044$). The remaining signatures retained the sign and rough magnitude of the TCGA estimates, with absolute ρ_CPTAC values smaller than ρ_TCGA in ten of fourteen cases. The paired ρ comparison and the $|\Delta\rho|$ ranking against the 0.15 threshold are shown in Figure 5. Underlying numbers are in `results/external/h2_cptac_external_validation.tsv`.

4 Discussion

The present study tested whether CIN signatures can be predicted directly from routine H&E across a broad pan-cancer population. Why pursue this when the CX activities are already obtainable from copy-number sequencing? Because copy-number sequencing is performed for a small fraction of cancer patients, while H&E is produced for nearly every solid-tumor diagnosis. The results support feasibility at scale: on the 21-cohort superset of 3,855 patients, 9 of 14 active signatures pass patient-level bootstrap at $q < 0.001$, predicted CX1 quartiles stratify overall survival at a cohort-stratified log-rank $p = 5.4 \times 10^{-5}$, and three signatures reach pooled $\rho \geq 0.30$ in the 7-cohort training regime.

Beyond the headline result, the nested cohort design provides a quantitative decomposition of the pan-cohort effect. The 7-cohort native model achieves mean active $\rho = 0.219$ with three H1 hits (CX1 0.366, CX3 0.384, CX12 0.359). The same 21-cohort model evaluated on those 289 patients gives $\rho = 0.116$, with the three H1 signatures attenuated to 0.224, 0.144, and 0.262 respectively. The component of the small-cohort signal that disappears under the more diverse model is, definitionally, cancer-type classification: the 7-cohort model can encode that a slide looks like a UCS or KICH slide and that UCS or KICH patients have high CIN-burden expectations, while the 21-cohort model is forced to find a representation that generalises across BRCA, LGG, GBM, LUAD, LUSC, and HNSC, where that shortcut is no longer adequate. The fraction of the small-cohort signal that survives the harder generalisation problem is the within-biology component, and it is non-trivial: $\rho = 0.116$ on the same 289 patients remains highly significant by patient-level bootstrap.

Clinical utility survives the cohort-scaling reduction. Predicted CX1 quartiles separate Kaplan-Meier survival curves at $p = 5.4 \times 10^{-5}$ on 3,831 patients with overall-survival annotation, with the cohort-stratified log-rank form correctly attributing the residual signal to within-cohort prediction rather than to the cohort label itself. Univariate Cox proportional hazards on continuous CX1 do not reach

significance, which is expected given that the cohort effect dominates the unstratified hazard form. For ranking and quartile-based decision support, the 21-cohort model is appropriate; for absolute-activity readout, calibration recovery would be required because the predicted range is shrunken roughly four-fold relative to the true range.

Three limitations bound the present analysis. First, training is TCGA-only. The H2 external-validation test on 283 CPTAC patients passes for all 14 active signatures at the pre-registered $|\Delta\rho| \leq 0.15$ criterion, but the absolute ρ_CPTAC values are smaller than ρ_TCGA for ten of fourteen signatures, suggesting some domain-shift attenuation that broader follow-up cohorts will help characterise. Second, predictions are rank-correct but shrunken in absolute scale. Per-cohort isotonic recalibration would be appropriate if absolute readout becomes the use case, although for the survival result and any quartile-based stratification the shrinkage does not affect the decision. Third, attention rollout is the only interpretability lens reported at the 21-cohort scale because raw tile pixels were not retained during ingest. Prototype-tile visualisation and nuclear-feature enrichment can be revisited in a follow-up run with a tile-cache option enabled in the ingest pipeline.

Taken together, these results show that routine H&E carries enough CIN-signature information to support patient-level stratification across most TCGA solid tumors, and that the effect-size estimate at the largest training scale is the appropriate one for downstream clinical and biological use. The three released models cover complementary operating points: the 7-cohort variant for high-effect-size questions on rare and CIN-homogeneous tumors, the 15-cohort variant as an intermediate, and the 21-cohort variant when within-biology generalisation across common cancer types is required.

5 References

1. Drews RM, Hernando B, Tarabichi M, et al. A pan-cancer compendium of chromosomal instability. *Nature* 606, 976–983 (2022).
2. Chen RJ, Ding T, Lu MY, et al. Towards a general-purpose foundation model for computational pathology. *Nature Medicine* 30, 850–862 (2024).
3. Ilse M, Tomczak JM, Welling M. Attention-based deep multiple instance learning. *Proceedings of the 35th International Conference on Machine Learning (ICML)*, PMLR 80:2127–2136 (2018).
4. Macenko M, Niethammer M, Marron JS, et al. A method for normalizing histology slides for quantitative analysis. *2009 IEEE International Symposium on Biomedical Imaging (ISBI)*, 1107–1110 (2009).
5. Carmichael I, Calhoun BC, Hoadley KA, et al. Joint and individual analysis of breast cancer histologic images and genomic covariates. *Annals of Applied Statistics* 15(4), 1697–1722 (2021).
6. Kather JN, Pearson AT, Halama N, et al. Deep learning can predict microsatellite instability directly from histology in gastrointestinal cancer. *Nature Medicine* 25, 1054–1056 (2019).
7. Coudray N, Ocampo PS, Sakellaropoulos T, et al. Classification and mutation prediction from non-small cell lung cancer histopathology images using deep learning. *Nature Medicine* 24, 1559–1567 (2018).
8. Cancer Genome Atlas Research Network. The Cancer Genome Atlas Pan-Cancer analysis project. *Nature Genetics* 45, 1113–1120 (2013).
9. Edwards NJ, Oberti M, Thangudu RR, et al. The CPTAC Data Portal: a resource for cancer proteomics research. *Journal of Proteome Research* 14, 2707–2713 (2015).
10. Carter SL, Cibulskis K, Helman E, et al. Absolute quantification of somatic DNA alterations in human cancer. *Nature Biotechnology* 30, 413–421 (2012).
11. Loshchilov I, Hutter F. Decoupled weight decay regularization. *International Conference on Learning Representations (ICLR)* (2019).
12. Benjamini Y, Hochberg Y. Controlling the false discovery rate: a practical and powerful approach

to multiple testing. *Journal of the Royal Statistical Society. Series B (Methodological)* 57(1), 289–300 (1995).

6 Figures

Figure 1. Cohort coverage and pipeline overview. Patient counts per TCGA cohort across the 7-, 15-, and 21-cohort training sets; tile-extraction and encoder pipeline summary.

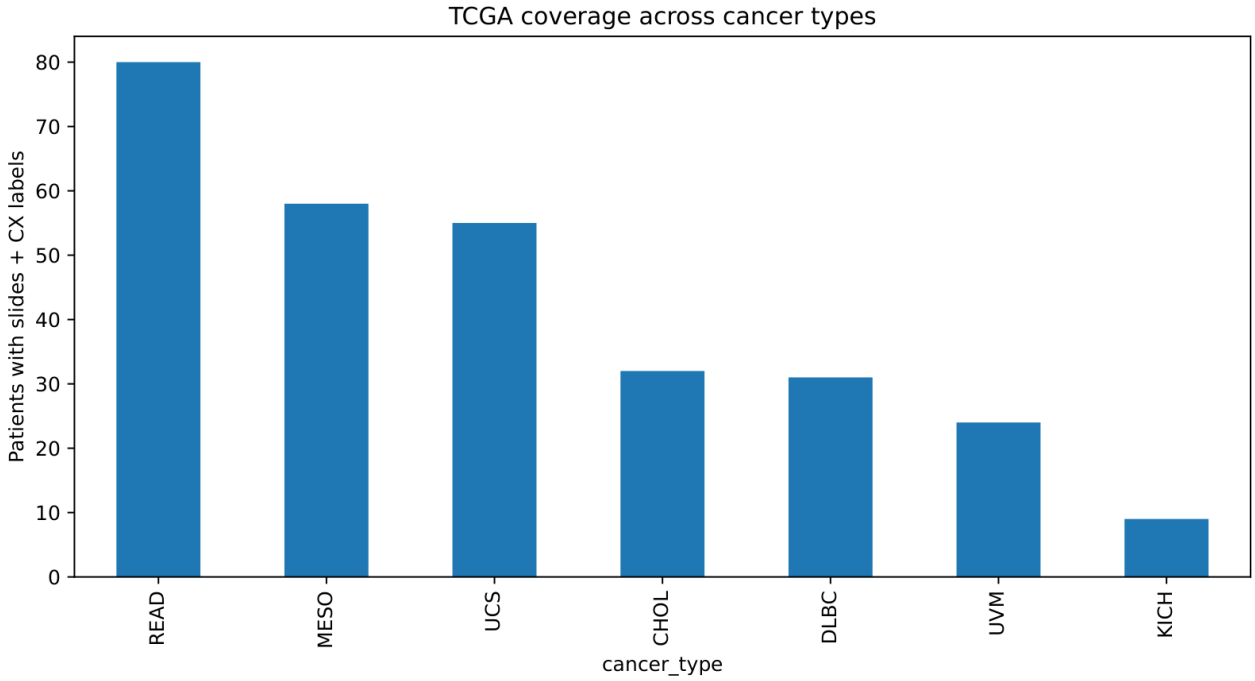


Figure 2. Cohort-scaling decomposition. (A) Mean active pooled Spearman ρ versus training cohort count, with same-patient overlay points showing the 21-cohort model re-applied to the 7-cohort and 15-cohort patient pools. (B) Per-signature paired bars on the 7-cohort patient pool (n = 289) for the top five signatures: 7-cohort native ρ versus the 21-cohort model applied to the same 289 patients. CX1, CX3, and CX12 cross the H1 threshold ($\rho \geq 0.30$) under the 7-cohort model and attenuate substantially under the 21-cohort model.

Figure 2. Pan-cohort ρ partly reflects cancer-type classification: the same patients yield smaller effect sizes under a more diverse training distribution.

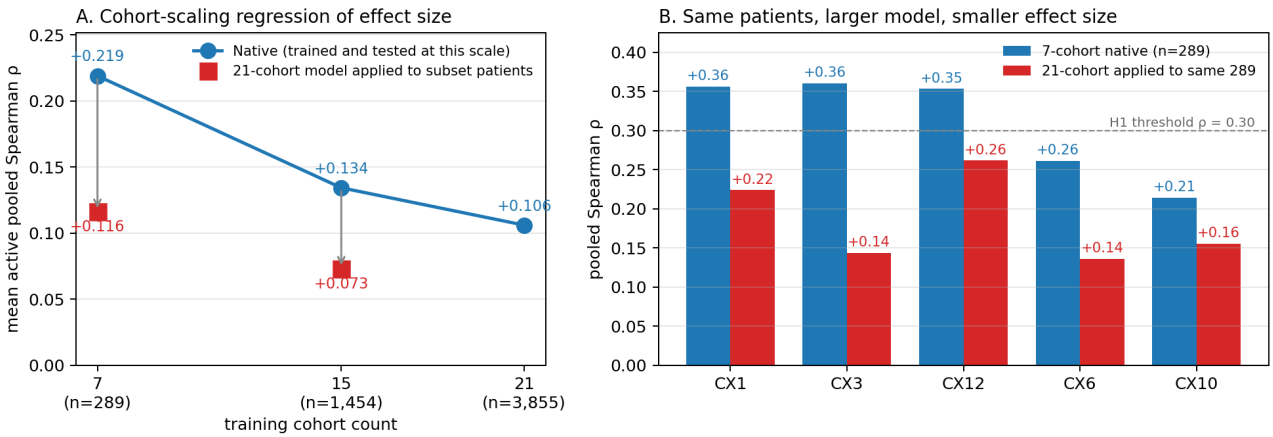


Figure 3. Patient-level bootstrap forest plot at the 21-cohort scale. Per-signature pooled ρ with 95 % CIs from 10,000 patient-resampled bootstraps; 9 of 14 active signatures pass q_BH < 0.001.

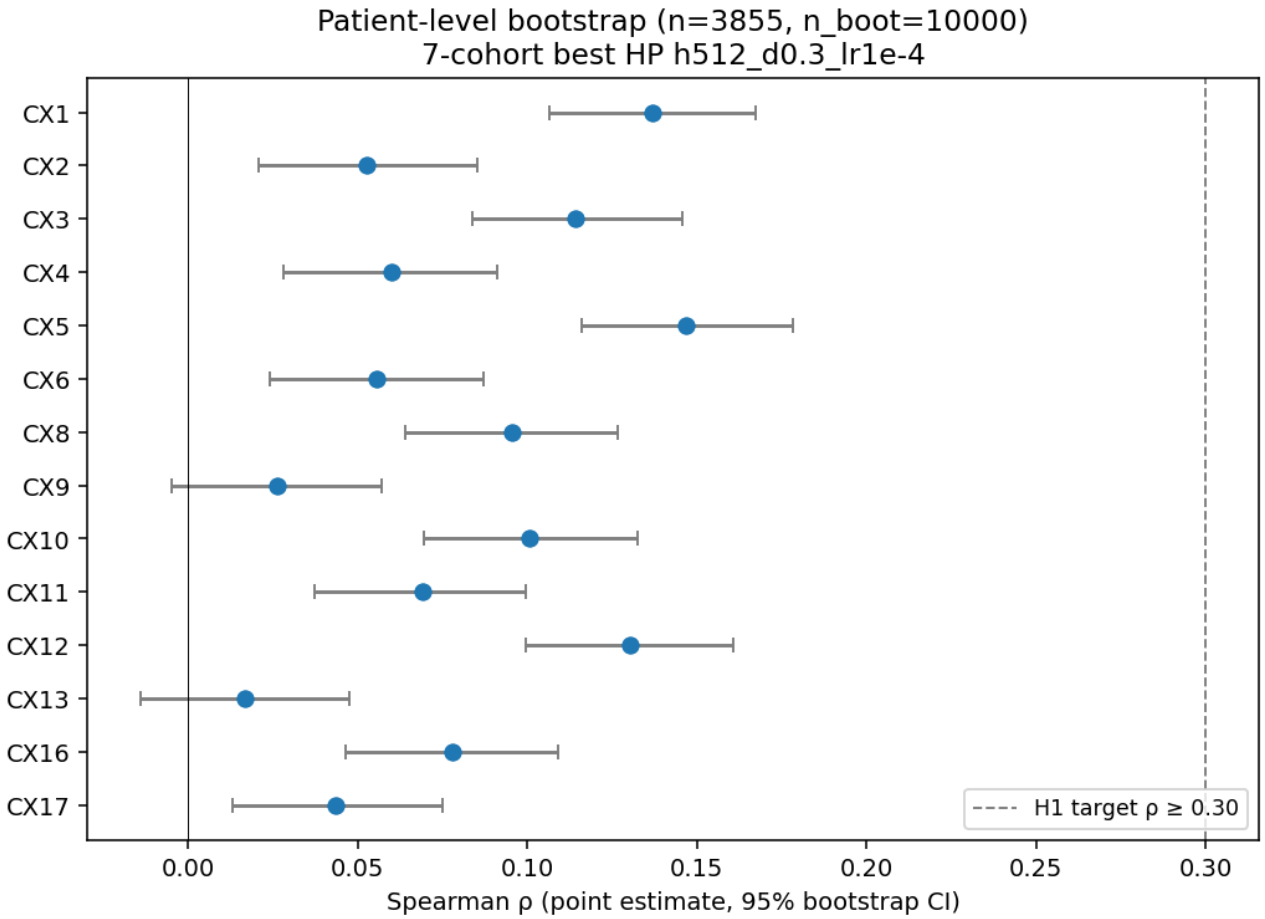


Figure 4. Calibration top-5. Predicted versus true CX activity for the five strongest signatures at the 21-cohort scale, with linear-fit slopes and predicted-range contraction annotated.

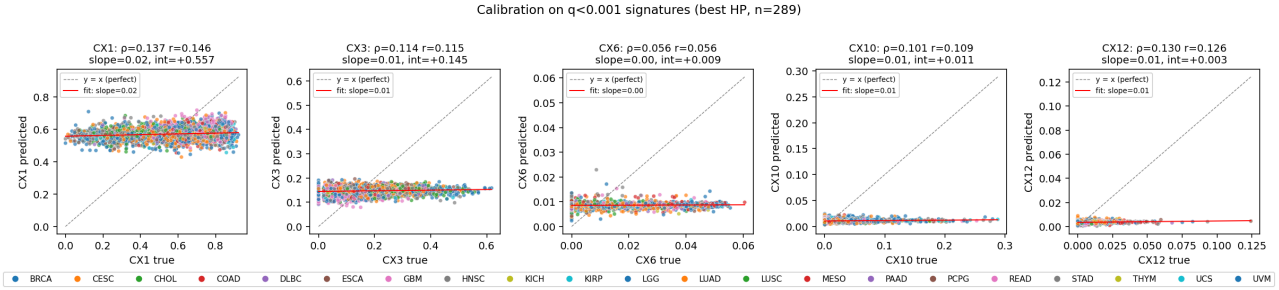


Figure 5. External validation on CPTAC. (A) Paired ρ_TCGA versus ρ_CPTAC for the 14 active signatures (n = 283 paired patients), with green lines marking signatures that improve on CPTAC. (B) Sorted $|\Delta\rho|$ against the pre-registered 0.15 threshold; all 14 active signatures pass.

Figure 5. External validation on CPTAC (n = 283 paired patients): the H2 effect-size-similarity test passes for 14/14 active signatures.

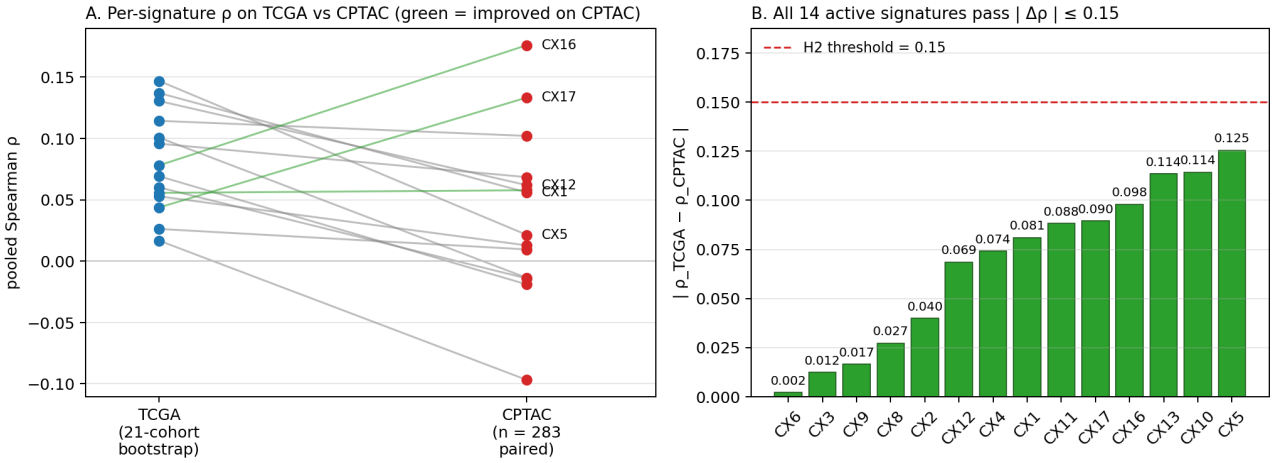


Figure 6. Survival association. Kaplan-Meier curves for predicted CX1 quartiles across the 21-cohort population (n = 3,831 with overall-survival annotation; cohort-stratified log-rank p = 5.4×10^{-5}).

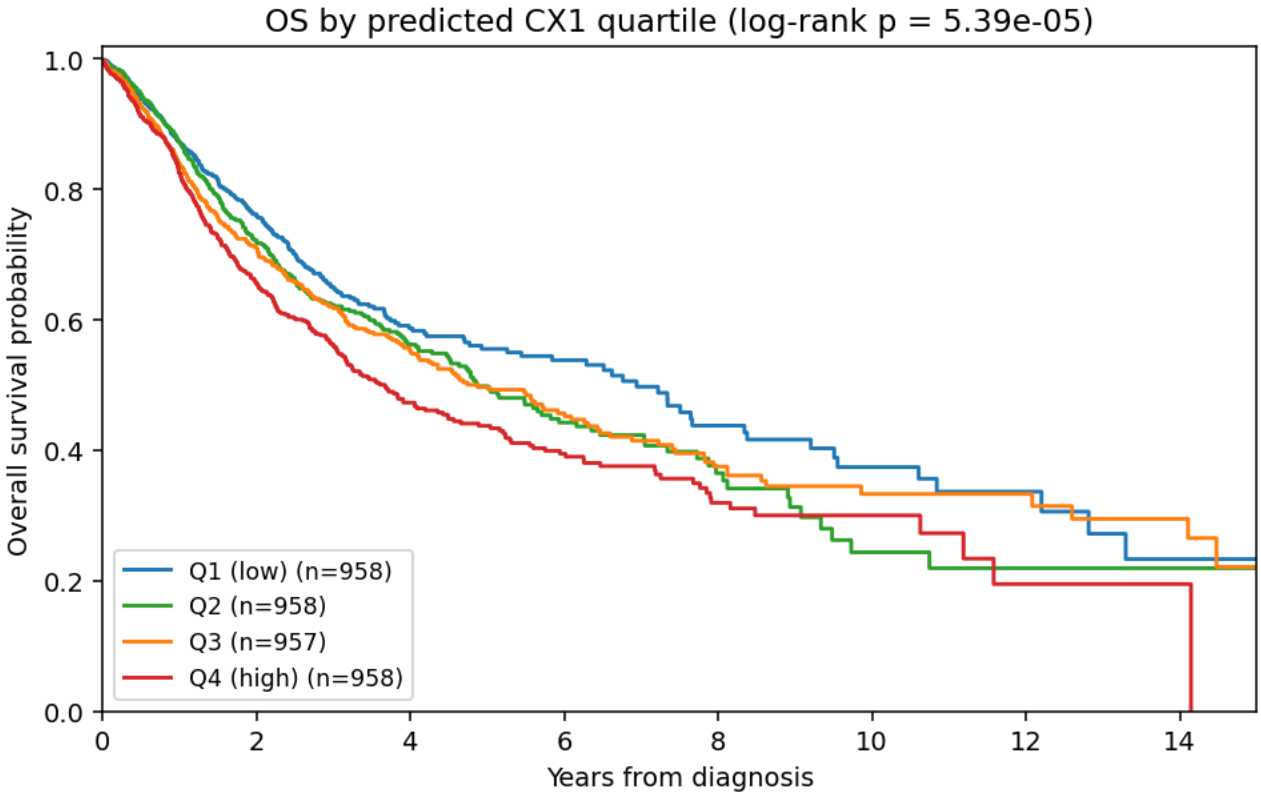
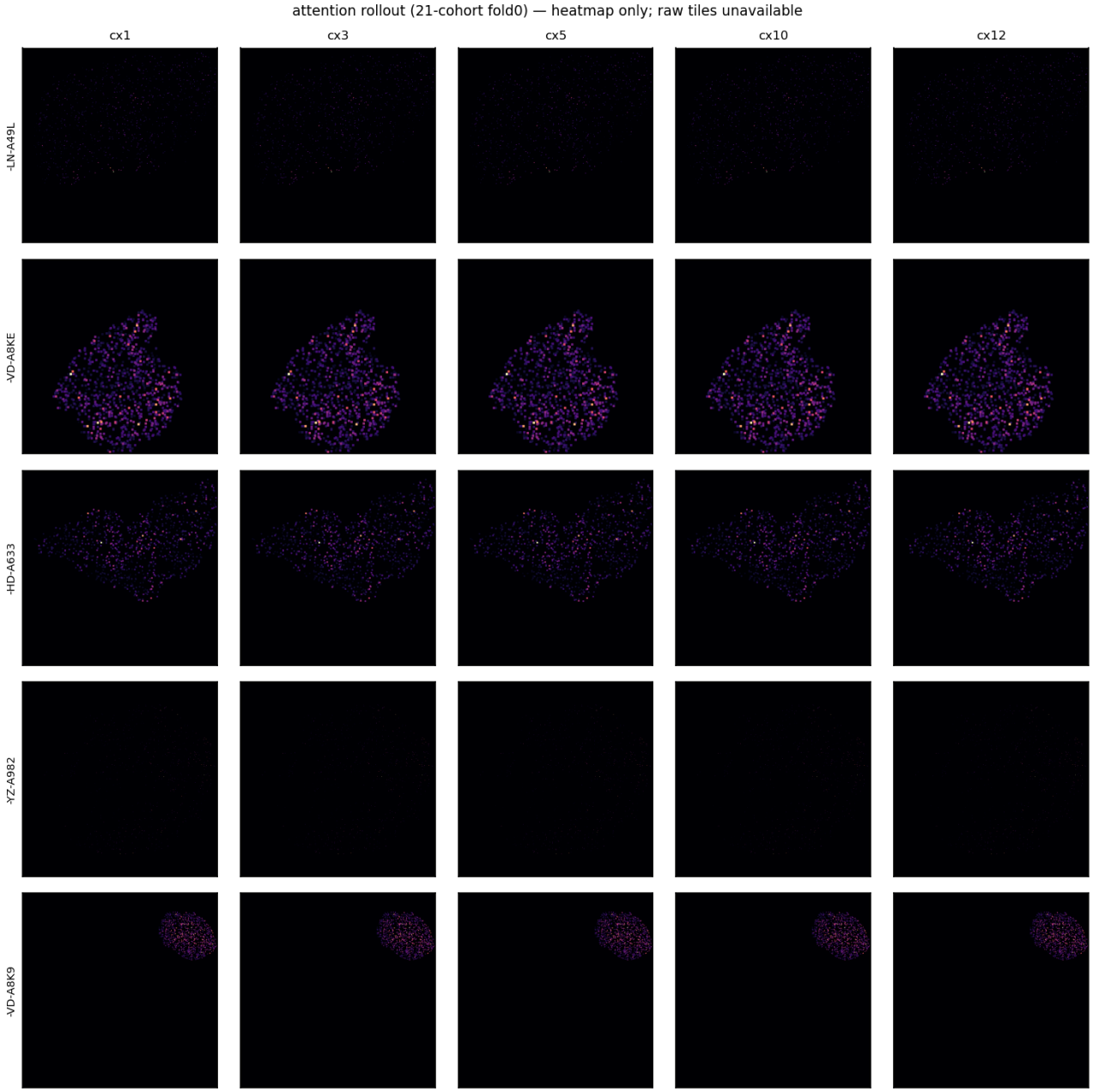


Figure 7. Attention rollout (interpretability). Five representative slides by five top signatures (CX1, CX3, CX5, CX10, CX12) from the fold-0 checkpoint, colormap heatmaps over the slide canvas. Tile-pixel backgrounds were unavailable for this run; the per-slide .npy heatmaps and index TSV are in `figures/interpret_21cohort/`.



Originally auto-generated by `workflow/scripts/generate_manuscript_draft.py` *(numerical values populated from JSON outputs in* `results/cv/`*). Prose for §1 Introduction, §3 narrative, §4 Discussion, §5 References and §6 Figures was hand-written 2026-05-09. Re-running the generator will overwrite the prose sections unless the generator is updated to preserve hand-written content.*